

# TrES-5b

R-1895

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-5\_160808 · created 2026-08-03T13:24:23+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2457608.8125 +/- 0.0022 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.185 +/- 0.053                 |
| Transit depth (Rp/Rs)^2                           | 3.42 +/- 1.97 [%]               |
| Orbital Inclination (inc)                         | 89.48 +/- 2.85                  |
| Airmass coefficient 1 (a1)                        | 1.1 +/- 0.13                    |
| Airmass coefficient 2 (a2)                        | -0.08 +/- 0.15                  |
| Scatter in the residuals of the lightcurve fit is | 11.82 %                         |
| Best Comparison Star                              | #2 - [346, 259]                 |
| Optimal Aperture                                  | 2.69                            |
| Optimal Annulus                                   | 5.52                            |
| Transit Duration (day)                            | 0.0875 +/- 0.0068               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                       |
|--------------------------------------|-------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.185 $\pm$ 0.053 vs 0.142 (deviation 0.71 $\sigma$ ) |
| Duration fit vs published            | 0.0875 d vs 0.0758 d (deviation 1.49 $\sigma$ )       |
| Mid-time O-C                         | -4.4 min                                              |
| Depth SNR                            | 3.0                                                   |
| Residual scatter                     | 11.82 %                                               |

## Light curve

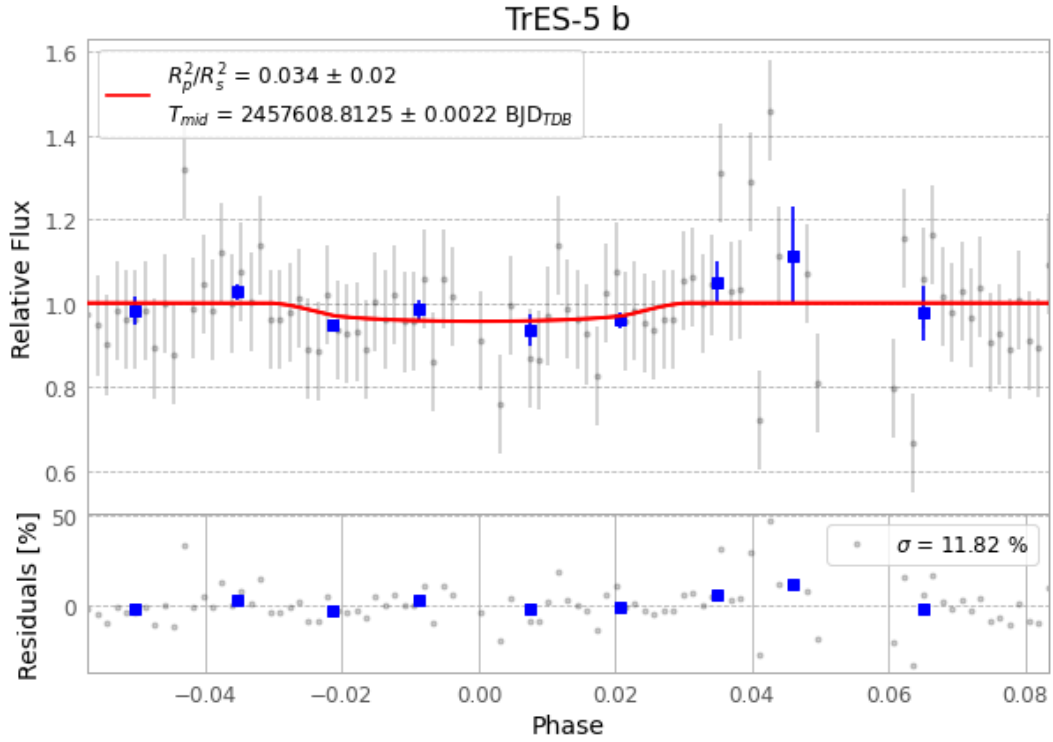


Figure 1 — FinalLightCurve\_TrES-5 b\_2016-08-07.png (SHA-256 20b7d73e8b2b7b62...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [441, 225], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 72490042ca873d12...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

101 FITS frames (66 MB), TRES-5160808052413.FITS → TRES-5160808102412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-160808.log (ca79b1024e87...), FinalLightCurve\_TrES-5 b\_2016-08-07.png (20b7d73e8b2b...), FinalLightCurve\_TrES-5 b\_2016-08-07.csv (a89630b84640...)

## Compute resources

Wall time 220.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

---

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 13:24Z.